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COMPUTER NETWORK
ASSIGNMENT-I

6.6.26

1. Hospital Communication System:

a. which transmission medium should be selected?

optical fibre cable should be selected

b. what characteristics make it suitable?

• Immune to electromagnetic interference: Medical equipment such as MRI machines generate strong electrogmagnetic fields. Fiber Optics use lights signals instead of electrical signals, so they are unaffected by interferene.

• High Bandwidth: can transmit large medical image or patient data quickly.

• Long distance Communication: Supports communication over long distance without significant signal los.

• High Security: Data is difficult to tap because signal travel as light through glass filters.

c. What risks arise if the wrong medium is used?

• If copper cables (UTP/STP) are used, Signal Interference from MRI machine May corrupt data.

• Network errors and communication failures may occur.

• Pateint monitoring data may be depend or lost

• Reduce reliability can affect healthcare operations and pateint safety.

2 MULTINATIONAL COOPORATION DATA CENTERS:-

a. choice of transmission medium:

fiber Optic communication is the preferred medium.

b. security and bandwidth requirements

security: Supports encrypted communication (VPN, SSL, TLS). Difficult to Intercept compared to wireless transmission. Suitable for sensitive corporate data.

Bandwidth: Very high bandwidth. Supports cloud services of Backups, vedio, configuration and data syncronization.

c. Reliability configuration:

Low signal attenuation over long distance High resistance to environmental noise Redundant fiber paths can provide backup connectivity.

3. college Computer Laboratory (Wired LAN)

a. why is a wired LAN Suitable for this environment

- provide high speed data transfer
- stable and reliable connection
- Supports simultaneous by many users
- suitable for file sharing, printer sharing and Internet access
- More secure than wireless Networks

b. What role does the switch play in network?

=> It connects all computers in the LAN. Learns the MAC address of each device. Sends data only to the Intendended destination Port.

=> Reduce unnecessary network, traffic Improves network performance.

c. How does ethernet ensure reliable communication?

=> Uses standardized ethernet frames. Includes source and destination MAC address. Use frame check sequence for error detection. Detects corrupted frames and discards them.

=> supports full duplex communication, reducing collisions

d. Advantages and limitations of a wired LAN:

Advantages:

=> high speed

=> Reliable Communication

=> Better Security

=> Low interference

=> stable performance

Limitations:

⇒ cable installation cost

⇒ Less mobility

⇒ Difficult to expand physically.

⇒ cable maintainance required

⇒ more wiring needed.

4. WIFI COMMUNICATION (CSMA/CA)

a. Why can't wifi derives use collision detection like ethernet?

⇒ A device cannot transmit and listen at the same time. The transmitted signal is much stronger than incoming signals.

⇒ Therefore, collisions cannot be elected directly. hence wifi uses CSMA/CA instead of CSMA/CD.

b. How does CSMA/CA avoid collision before transmission?

⇒ Device listens to channel. If the channel is busy, it waits. When the channel becomes idle, it waits for a fixed period (IFS)

⇒ A random backoff timer is selected. The desires transmits only when the timer reaches zero.

c.) What is the purpose of the random backoff timer?

⇒ Prevents two devices from transmitting at same times.

⇒ Each devices waits for a random duration the devices whose timer expires first gets access to the medium.

d) How does the ACK mechanism improve reliability?

⇒ Sender transmit a frame

⇒ Receiver checks the frame

⇒ It recieved correctly, reciever sends an ACK.

⇒ If ACK is not recieved within a timeout period the sender retransmits the frame.